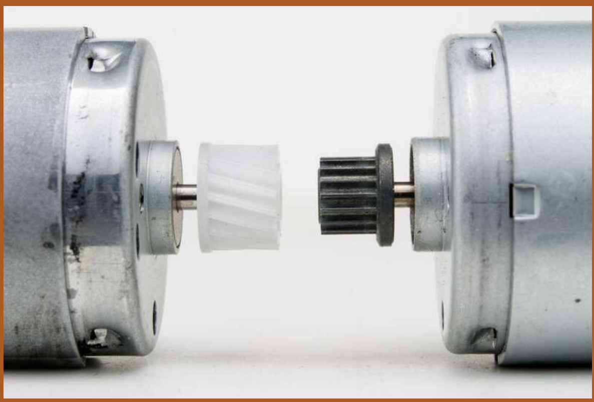
Worksheet 4

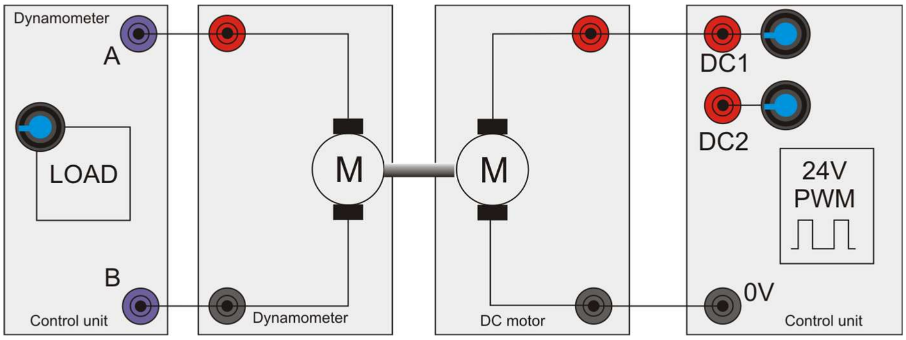
DC motor characteristics 1

Modern electrical machines



Different types of motor have different characteristics, such as top speed, torque at different speeds , voltage and current ratings, power output and efficiency.

Photo: these tiny DC motors are used in small handheld devices like toys.



Over to you:

1) Set up the circuit shown above. Connect the DC motor to the control unit's DC1 power supply V+ and 0V terminals.
2) Energise the DC motor to put pressure on the dynamometer load cell rather than the balance. (If necessary, reverse the power supply connections .)
3) Connect the dynamometer to the load terminals on the control unit to set a load across the dynamometer.
4) Connect the control box to your computer using the USB lead.
5) On the computer, load the 'Open Control DC' program. Press the 'PLAY' style button on the top left and then click on the 'RUN' button.
6) Set the PWM DC output to around 30%.
7) Give the DC motor a small load of around 50% on the dynamometer.
8) Investigate the effect of varying the PWM voltage between 30% and 100%, taking six readings across that range. In the table on the next page record the speed, torque, and Driver Power coming from the power supply.
9) Use the information in the 'Key Relationships' section on page 4 to calculate the mechanical power delivered to the dynamometer.
10) In addition, calculate the efficiency of the motor at each of the six settings .
11) On the graph paper provided, use these readings to plot a graph to show how the torque (y-axis) varies with speed (x-axis).

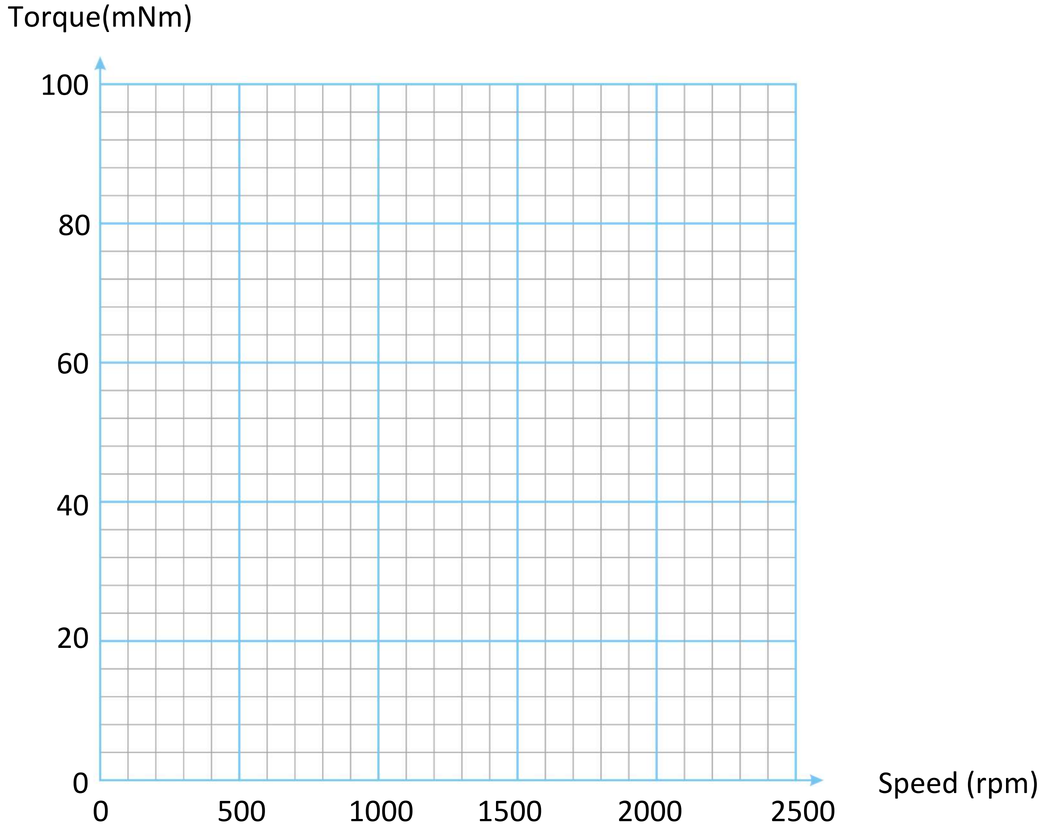
Note that for repeated calculations you will find it easier to use a spreadsheet.

Worksheet 4

DC motor characteristics 1

Modern electrical machines

Voltage	Speed	Torque	Electrical power	Mechanical power	Efficiency



Variation of torque with speed

So what?

Each motor has characteristics that make it suitable for particular applications in industry. In this learning package, you meet a variety of motor types and learn to appreciate their characteristics.

Worksheet 4

DC motor characteristics 1

Over to you

Describe some of the characteristics of the DC motor.

What speeds give maximum and minimum torque?

At what speed is it most efficient?

When would you use a DC motor as opposed to a different type?